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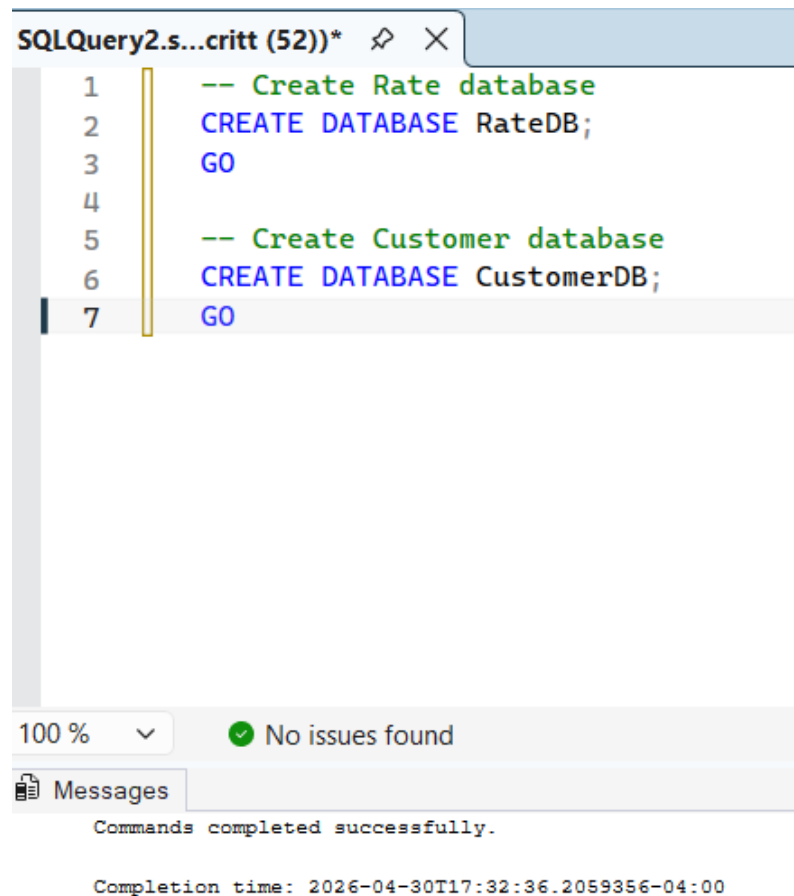
ISIN-325

4/30/2026

SQL Assignment 4

1) Define and populate Rate and Customer tables in two separate databases. Suggested that the Rate table be in a Rate database and the Customer table be in a Customer database.

Create database



The screenshot shows a SQL query window titled "SQLQuery2.s...critt (52))*". The query contains two statements to create databases. The first statement creates a database named "RateDB" and the second creates a database named "CustomerDB". Both statements are preceded by a green comment line. The query is executed successfully, as indicated by the "Messages" pane at the bottom, which shows "Commands completed successfully." and a completion time of "2026-04-30T17:32:36.2059356-04:00".

```
1  -- Create Rate database
2  CREATE DATABASE RateDB;
3  GO
4
5  -- Create Customer database
6  CREATE DATABASE CustomerDB;
7  GO
```

100 % ✓ No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-30T17:32:36.2059356-04:00

2) Include BeginAge and EndAge columns for the Rate table. The age ranges go in 5 year increments up to age 99. Separate records for male and female.

Create tables

1. Rate table

```
SQLQuery2.sql...critt (52))* ✕ X
1  USE RateDB;
2  GO
3
4  -- Create Rate table with required fields
5  CREATE TABLE Rate(
6      RateID INT IDENTITY(1,1) PRIMARY KEY,
7      BeginAge INT NOT NULL,
8      EndAge INT NOT NULL,
9      Gender CHAR(1) NOT NULL, -- M or F
10     Rate DECIMAL(9,2) NOT NULL
11 );
```

100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-30T17:33:36.9032362-04:00

2. Customer Table

```
SQLQuery3.s...critt (52))* ✕ X
1  USE CustomerDB;
2  GO
3
4  -- Create Customer table with Gender
5  CREATE TABLE Customer(
6      CustomerID INT IDENTITY(1,1) PRIMARY KEY,
7      FName NVARCHAR(50),
8      LName NVARCHAR(50),
9      DOB DATE,
10     Gender CHAR(1)
11 );
```

100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-30T17:39:55.1154407-04:00

Populate Tables

1. Rate table

```
SQLQuery2.s...critt (52))* ✕ ×
1  USE RateDB;
2  GO
3
4  -- Insert all age ranges (Male + Female)
5  INSERT INTO Rate (BeginAge, EndAge, Gender, Rate) VALUES
6  -- 0-4
7  (0,4,'M',15.00),(0,4,'F',14.00),
8  -- 5-9
9  (5,9,'M',18.00),(5,9,'F',17.00),
10 -- 10-14
11 (10,14,'M',20.00),(10,14,'F',19.00),
12 -- 15-19
13 (15,19,'M',25.00),(15,19,'F',23.00),
14 -- 20-24
15 (20,24,'M',30.00),(20,24,'F',28.00),
16 -- 25-29
17 (25,29,'M',33.00),(25,29,'F',31.00).
100 % ▾ No issues found
Messages
(40 rows affected)
Completion time: 2026-04-30T17:37:46.7448005-04:00
```

2. Customer Table

```
SQLQuery3.s...critt (52))* ✕ ×
SQLQuery3.sql - ETHANSLAPTOP\SQLEXPRESS.CustomerDB (ETHANSLAPTOP\critt (52))*
2  GO
3
4  INSERT INTO Customer (FName, LName, DOB, Gender) --Insert into customer table
5  VALUES
6  ('John','Smith','1998-05-10','M'),
7  ('Emily','Jones','2005-03-22','F'),
8  ('Michael','Brown','1980-07-15','M'),
9  ('Sarah','Davis','1975-11-02','F'),
10 ('Chris','Wilson','2015-06-18','M'),
11 ('Anna','Taylor','2010-09-25','F'),
12 ('David','Moore','1965-01-30','M'),
13 ('Laura','White','1990-12-12','F'),
14 ('James','Hall','1988-04-03','M'),
15 ('Olivia','Allen','2002-08-17','F'),
16
17 ('Daniel','Yound','1970-09-09','M').
100 % ▾ 5 0 ⬆ ⬇
Messages
(48 rows affected)
Completion time: 2026-04-30T17:41:47.1105731-04:00
```

Proof of Population

1. Rate Table

SQLQuery2.s...critt (52))*
1 SELECT * FROM Rate;

100 %
No issues found

	RateID	BeginAge	EndAge	Gender	Rate
1	1	0	4	M	15.00
2	2	0	4	F	14.00
3	3	5	9	M	18.00
4	4	5	9	F	17.00
5	5	10	14	M	20.00
6	6	10	14	F	19.00
7	7	15	19	M	25.00
8	8	15	19	F	23.00
9	9	20	24	M	30.00
10	10	20	24	F	28.00
11	11	25	29	M	33.00
12	12	25	29	F	31.00
13	13	30	34	M	36.00
14	14	30	34	F	34.00
15	15	35	39	M	39.00
16	16	35	39	F	37.00
17	17	40	44	M	42.00
18	18	40	44	F	40.00
19	19	45	49	M	45.00
20	20	45	49	F	43.00
21	21	50	54	M	48.00
22	22	50	54	F	46.00
23	23	55	59	M	52.00
24	24	55	59	F	50.00
25	25	60	64	M	60.00
26	26	60	64	F	58.00
27	27	65	69	M	70.00
28	28	65	69	F	68.00
29	29	70	74	M	85.00
30	30	70	74	F	82.00
31	31	75	79	M	95.00

2. Customer Table

SQLQuery3.s...critt (52))*
1 SELECT * FROM Customer;

100 %
1 0

	CustomerID	FName	LName	DOB	Gender
1	1	John	Smith	1998-05-10	M
2	2	Emily	Jones	2005-03-22	F
3	3	Michael	Brown	1980-07-15	M
4	4	Sarah	Davis	1975-11-02	F
5	5	Chris	Wilson	2015-06-18	M
6	6	Anna	Taylor	2010-09-25	F
7	7	David	Moore	1965-01-30	M
8	8	Laura	White	1990-12-12	F
9	9	James	Hall	1988-04-03	M
10	10	Olivia	Allen	2002-08-17	F
11	11	Daniel	Young	1970-09-09	M
12	12	Sophia	Hernandez	1999-01-25	F
13	13	Matthew	King	1985-03-11	M
14	14	Ava	Wright	2007-07-07	F
15	15	Andrew	Lopez	1960-06-14	M
16	16	Mia	Hill	2012-11-21	F
17	17	Joshua	Scott	1955-02-02	M
18	18	Charlot...	Green	1983-10-30	F
19	19	Ryan	Adams	1993-12-19	M
20	20	Amelia	Baker	2001-05-05	F
21	21	Brandon	Nelson	1978-06-01	M
22	22	Evelyn	Carter	1968-08-23	F
23	23	Justin	Mitchell	1996-09-09	M
24	24	Abigail	Perez	2016-04-12	F
25	25	Tyler	Roberts	1982-01-18	M
26	26	Harper	Turner	2008-03-27	F
27	27	Kevin	Phillips	1973-07-07	M
28	28	Ella	Campbell	1994-10-10	F
29	29	Jason	Parker	1987-11-11	M
30	30	Scarlett	Evans	2003-02-02	F
31	31	Anna	Edwards	1983-12-12	M

3) Create SQL routines that perform the following:

Add Insurance Rates - assign people an insurance rate based on birthdate (age) and gender.

* example: Someone born May 5, 2000 would be 26 years old, so they would be using the rates from 25 to 29 in the rate table.

* this is a SQL routine, not another column in a table. You compute the assigned rate.

SQLQuery4.s...critt (51)*

```
1 SELECT
2     C.FName,
3     C.LName,
4     C.DOB,
5
6     -- Calculate Age
7     DATEDIFF(YEAR, C.DOB, GETDATE()) -
8     CASE
9         WHEN MONTH(C.DOB) > MONTH(GETDATE())
10          OR (MONTH(C.DOB) = MONTH(GETDATE()) AND DAY(C.DOB) > DAY(GETDATE()))
11          THEN 1
12          ELSE 0
13     END AS Age,
14
15     R.Rate
16
17 FROM CustomerDB.dbo.Customer C
18 JOIN RateDB.dbo.Rate R
19     ON C.Gender = R.Gender
20    AND (
21        DATEDIFF(YEAR, C.DOB, GETDATE()) -
22        CASE
23            WHEN MONTH(C.DOB) > MONTH(GETDATE())
24             OR (MONTH(C.DOB) = MONTH(GETDATE()) AND DAY(C.DOB) > DAY(GETDATE()))
25            THEN 1 ELSE 0
26        END
27    BETWEEN 0 AND 13)
```

70 % No issues found

	FName	LName	DOB	Age	Rate
1	John	Smith	1998-05-10	27	33.00
2	Emily	Jones	2005-03-22	21	28.00
3	Michael	Brown	1980-07-15	45	45.00
4	Sarah	Davis	1975-11-02	50	46.00
5	Chris	Wilson	2015-06-18	10	20.00
6	Anna	Taylor	2010-09-25	15	23.00
7	David	Moore	1965-01-30	61	60.00
8	Laura	White	1990-12-12	35	37.00
9	James	Hall	1988-04-03	38	39.00
10	Olivia	Allen	2002-08-17	23	28.00
11	Daniel	Young	1970-09-09	55	52.00
12	Sophia	Hernandez	1999-01-25	27	31.00
13	Matthew	King	1985-03-11	41	42.00

Add Rider - Alter Customer table to include Rider flag. Compute and display rider cost - rider will increase rate by 1.5.

```
SQLQuery4.s...critt (51))* ✕
1  -- Add Rider column
2  ALTER TABLE CustomerDB.dbo.Customer
3  ADD Rider BIT DEFAULT 0;
```

100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-30T17:49:51.8205523-04:00

-Set Rider

```
SQLQuery4.s...critt (51))* ✕
1  -- Set Rider = 1 for 10 customers
2  UPDATE CustomerDB.dbo.Customer
3  SET Rider = 1
4  WHERE CustomerID <= 10;
```

100 % 1 0

Messages

(10 rows affected)

Completion time: 2026-04-30T17:51:13.5378770-04:00

-Calculate final rate with Rider

SQLQuery4.s...critt (51))*

```

1  SELECT -- Select customer's first and last name
2  C.FName,
3  C.LName,
4
5  R.Rate AS BaseRate, -- Display the original rate from the Rate table
6
7  CASE
8  WHEN C.Rider = 1 THEN R.Rate * 1.5 -- If Rider = 1 → increase rate by 1.
9  ELSE R.Rate -- If Rider = 0 → keep original rate
10 END AS FinalRate
11
12 FROM CustomerDB.dbo.Customer C -- From Customer table in CustomerDB
13
14 JOIN RateDB.dbo.Rate R -- Join with Rate table in RateDB
15 ON C.Gender = R.Gender -- Match customer gender to rate gender
16
17 AND (
18 -- Calculate customer's age
19 DATEDIFF(YEAR, C.DOB, GETDATE()) -
20 CASE
21 -- Adjust age if birthday hasn't occurred yet this year
22 WHEN MONTH(C.DOB) > MONTH(GETDATE())
23 OR (MONTH(C.DOB) = MONTH(GETDATE()) AND DAY(C.DOB) > DAY(GETDATE()))
24 THEN 1
25 ELSE 0
26 END
27 ) BETWEEN R.BeginAge AND R.EndAge; -- Match to correct rate range

```

70 % 1 0

Results Messages

	FName	LName	BaseRate	FinalRate
1	John	Smith	33.00	49.500
2	Emily	Jones	28.00	42.000
3	Michael	Brown	45.00	67.500
4	Sarah	Davis	46.00	69.000
5	Chris	Wilson	20.00	30.000
6	Anna	Taylor	23.00	34.500
7	David	Moore	60.00	90.000
8	Laura	White	37.00	55.500
9	James	Hall	39.00	58.500
10	Olivia	Allen	28.00	42.000
11	Daniel	Young	52.00	52.000
12	Sophia	Hernandez	31.00	31.000
13	Matthew	King	42.00	42.000

- Report of all customers sorted by age range, LName, FName. Displays the Rate/Rate x Rider (whichever applies - join)

Customer: CustomerID, FName, LName, DOB, Gender (then alter to add Rider flag to any 10 of your customers) (50 people, spread across ranges)

Rate: RateID, BeginAge, EndAge, Gender, Rate (Age in increments of 5 years, 0-4, 5-9, 10-14... to 99) (Rate - dollar amount, different (increases) in each age increment)

SQLQuery4.s...critt (51))*

```

1  SELECT
2      -- Customer name
3      C.FName,
4      C.LName,
5
6      -- Display the age range from the Rate table
7      R.BeginAge,
8      R.EndAge,
9
10     -- Calculate final rate with Rider
11     CASE
12         WHEN C.Rider = 1 THEN R.Rate * 1.5 -- Apply 1.5x if Rider exists
13         ELSE R.Rate -- Otherwise use base rate
14     END AS FinalRate
15
16 -- From Customer table
17 FROM CustomerDB.dbo.Customer C
18
19 -- Join with Rate table
20 JOIN RateDB.dbo.Rate R
21     ON C.Gender = R.Gender -- Ensure gender matches correct rate
22
23     -- Match customer's calculated age to the correct range
24     AND (
25         DATEDIFF(YEAR, C.DOB, GETDATE()) -
26         CASE
27             -- Adjust age if birthday has not occurred yet this year
28             WHEN MONTH(C.DOB) > MONTH(GETDATE())
29             OR (MONTH(C.DOB) = MONTH(GETDATE()) AND DAY(C.DOB) > DAY(GETDATE()))
30             THEN 1
31             ELSE 0
32         END
33     ) BETWEEN R.BeginAge AND R.EndAge
34
35 ORDER BY
36     R.BeginAge, -- Sort by age range first
37     C.LName, -- Then by last name
38     C.FName; -- Then by first name

```

70%

1

0

↑

↓

Results

Messages

	FName	LName	BeginAge	EndAge	FinalRate
1	Grace	Collins	5	9	17.000
2	Lily	Cook	10	14	19.000
3	Addison	Cox	10	14	19.000
4	Mia	Hill	10	14	19.000
5	Abigail	Perez	10	14	19.000